Notes 08/24 (Propositional Logic)

Proposition:

Statements that are either true or false.

Questions are not propositions. Demands are not propositions.

Example:

$x + y = 3$ and $x = 7$ are *not* propositions — they are ambiguous, since we don't know what x, y are.

Propositional variables denote propositions.

Negation. $\neg p \equiv$ not p.

Truth table:

p	$\neg p$
T	F
F	T

Conjunction of p and q: $p \wedge q$ (and).
Disjunction of p and q: $p \vee q$ (or).
Exclusive or (XOR): $p \oplus q \equiv (p \vee q) \wedge \neg(p \wedge q)$.

p	q	$p \oplus q$	$p \vee q$	$p \wedge q$
T	T	F	T	T
T	F	T	T	F
F	T	T	T	F
F	F	F	F	F

Conditional statement:

$p \to q$.
p is the **sufficient condition / hypothesis**; q is the **conclusion** or the **necessary condition**.

$$\neg(p \to q) \equiv p \wedge \neg q, \qquad p \to q \equiv \neg p \vee q.$$

Truth table:

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

Equivalent phrasings of $p \to q$:

1. If p, then q.
2. If p, q.
3. p is sufficient for q.
4. q if p.
5. q when p.
6. A necessary condition for p is q.
7. q unless $\neg p$.
8. p implies q.
9. p only if q.
10. a sufficient condition for q is p.
11. q whenever p.
12. q is necessary for p.
13. q follows from p.
14. q provided that p.

Related conditionals. Given $p \to q$:

- **Converse:** $q \to p$
- **Inverse:** $\neg p \to \neg q$
- **Contrapositive:** $\neg q \to \neg p$

Biconditionals — "if and only if", abbreviated "iff".

$$p \leftrightarrow q$$

Truth table:

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

p is a sufficient condition for q to happen, and q is a sufficient condition for p to happen.